

CROPCAST: THE SMART AGRI-COMMODITY PRICE PREDICTOR

**A PROJECT REPORT
(Phase I)**

Submitted by

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in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY
in

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



CHENNAI INSTITUTE OF TECHNOLOGY

(AUTONOMOUS)

(Affiliated to Anna University, Chennai & Recognized by AICTE New Delhi)
Chennai-600 069

SEPTEMBER-2026

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Certified that this project report (Phase – I)
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EXTERNAL EXAMINER

ACKNOWLEDGEMENT

We express our gratitude to our Chairman **Shri.P.SRIRAM** and all trust members of Chennai Institute of Technology for providing the facility and opportunity to do this project as a part of our undergraduate course.

We are grateful to our Principal **Dr.A.RAMESH M.E, Ph.D**, for providing us the facility and encouragement during the course of our work.

We would like to extend our thanks to our **Dean, Dr.V.SRINIVASA RAO M.Tech, PhD**, for providing their valuable suggestions throughout this project.

We sincerely thank our **Head of the Department, Ms. Shanmuga Sundari**, Assistant Professor, Department of Artificial Intelligence & Data Science, for having provided us with valuable guidance, resources and timely suggestions throughout our work.

We sincerely thank our **Project Supervisor, Dr.K.RAMANAN, M.Tech, Ph.D**, Associate Professor, Department of Artificial Intelligence & Data Science, for having provided us with valuable guidance, resources and timely suggestions throughout our work.

We would like to extend our thanks to our **Project Coordinator, Dr.K.RAMANAN M.Tech, Ph.D**, Associate Professor, Department of Artificial Intelligence & Data Science, for his valuable suggestions throughout this project.

We wish to extend our sincere thanks to all the **Faculty Members of the Department of Artificial Intelligence & Data Science**, for their valuable suggestions and their kind cooperation for the successful completion of our project.

We wish to acknowledge the help received from the **Lab Instructors of the Department of Artificial Intelligence & Data Science**, and others for providing valuable suggestions and for the successful completion of the project.

ABSTRACT

Fluctuation in agribusiness commodity prices causes unpredictable levels of stress for consumers, farmers, and policymakers across India, making price stability a persistent national concern. The Department of Consumer Affairs actively monitors 22 essential food commodities on a daily basis—including pulses such as gram, tur, urad, moong, and masur, and vegetables such as onion, potato, and tomato—through nearly 550 reporting centres nationwide. Despite this extensive monitoring mechanism, consistent control over price volatility remains challenging due to inherent seasonality of agricultural commodities, variations in crop yield, disruptions in supply chains, and complex market dynamics involving wages, inflation, exchange rates, and demand–supply imbalances. These factors collectively pose serious risks to food security, economic stability, and rural livelihoods. To address these challenges, this project proposes an AI–ML based predictive forecasting framework for essential pulses and vegetables, with a particular focus on onion, potato, and tomato. Unlike traditional statistical models such as ARIMA that rely primarily on historical price trends and linear assumptions, the proposed approach leverages machine learning and deep learning techniques to capture nonlinear relationships and temporal dependencies. The framework integrates multi-dimensional inputs including seasonality patterns, crop sowing cycles, production estimates, buffer stock releases, market intelligence indicators, and external factors such as weather conditions. Advanced data preprocessing and feature engineering techniques are employed to enhance data quality and model learning capability, followed by hybrid time-series modeling using Long Short-Term Memory (LSTM) networks and Random Forest algorithms to improve prediction accuracy and robustness. The resulting forecasts provide actionable insights that can assist policymakers in designing timely and effective market interventions, optimizing procurement strategies, regulating buffer stock releases, and administering subsidies more efficiently. By translating complex predictive outputs into interpretable and user-centric insights, the system bridges the gap between advanced analytics and real-world agricultural decision-making. Overall, this AI–ML driven initiative strengthens agricultural market intelligence, supports price stabilization efforts, enhances food security, and contributes to a more resilient, data-driven, and sustainable agricultural ecosystem aligned with the goals of Agriculture, Food Technology, and Rural Development in India.

Key Words:

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO
	ABSTRACT	i
	LIST OF FIGURES	ii
	LIST OF TABLES	iv
	SYMBOLS & ABBREVIATIONS	v
1.	INTRODUCTION	1
1.1	BACKGROUND	1
1.1.1	Global Context	2
1.1.2	Indian Context	2
1.2	PROBLEM STATEMENT	2
1.2.1	Our Understanding of the Problem	4
1.2.2	Planned Approach	5
1.3	AGRICULTURAL PRICE VOLATILITY	6
1.4	OBJECTIVES	6
1.4.1	General Objective	6
1.4.2	Specific Objectives	6
1.5	IMPORTANCE OF PRICE PREDICTION IN AGRICULTURE	9
1.6	ROLE OF AI AND ML IN PRICE PREDICTION	10
1.7	SIGNIFICANCE	11
1.8	SCOPE	11
1.9	LIMITATIONS OF THE EXISTING SYSTEM	11
1.10	OVERVIEW OF THE PROPOSED SYSTEM	12
2.	LITERATURE REVIEW	13
2.1	EXISTING RESEARCH	13
2.2	KEY OBSERVATIONS FROM EXISTING	16

	RESEARCH	
2.2.1	Traditional Statistical Models	16
2.2.2	Machine Learning Approaches	16
2.2.3	Deep Learning & Hybrid Models	16
2.2.4	Incorporation of External Factors	16
2.2.5	Decision Support & User-Centric Applications	16
2.3	GAPS IN EXISTING KNOWLEDGE	17
2.4	VALUE ADDITION OF THE PROPOSED SYSTEM	18
3.	METHODOLOGY	19
3.1	SYSTEM OVERVIEW	19
3.2	PROBLEM-SOLVING APPROACH	20
3.3	TECHNOLOGIES USED	20
3.3.1	Programming Language	20
3.3.2	Data Processing and Analytics Libraries	20
3.3.3	Machine Learning Frameworks	21
3.3.4	Deep Learning Frameworks	21
3.3.5	Forecasting Models	21
3.3.6	Web Application and Visualization Technologies	22
3.3.7	Data Storage and Management	22
3.3.8	Development Environment and Tools	22
3.3.9	Version Control	22
3.3.10	Evaluation Metrics and Tools	23
3.4	SYSTEM ARCHITECTURE	23
3.4.1	Data Sources Layer	24
3.4.2	Data Ingestion and Storage Layer	24
3.4.3	Processing and Analytics Layer	25
3.4.4	Forecasting Engine (AI/ML Layer)	25
3.4.5	Presentation Layer	26

3.4.6	System Integration and Workflow	26
3.4.7	Non-Functional Considerations	28
3.5	DATA COLLECTION	28
3.6	DATA PREPROCESSING	29
3.7	FEATURE ENGINEERING	29
3.8	FORECASTING MODELS AND EXPERIMENTAL SETUP	30
3.8.1	Random Forest (RF) Model	30
3.8.2	Long Short-Term Memory (LSTM) Model	30
3.8.3	Hybrid Models (LSTM + RF)	31
3.9	MODEL TRAINING AND EVALUATION	31
3.9.1	Data Splitting	31
3.9.2	Model Training	32
3.9.3	Model Evaluation	32
3.10	OUTPUT GENERATION AND VISUALIZATION	32
3.10.1	Web-Based Dashboard	32
3.10.2	Interactive Visualizations	32
3.10.3	Decision-Support Features	32
3.10.4	Usability and Interpretability	33
3.11	EXISTING SYSTEM AND PROPOSED WORK	33
3.11.1	Existing System	33
3.11.2	Proposed Work	33
7.	CONCLUSION	69
7.1	SUMMARY OF THE WORK	69
7.2	KEY CONTRIBUTIONS	70
7.3	ACHIEVEMENT OF OBJECTIVES	70
7.4	LIMITATIONS	71
7.5	FUTURE ENHANCEMENTS	71

REFERENCES	72
APPENDIX	74

LIST OF FIGURES

Figure No.	Title	Page No.
1.1	Influencing Factors for Price Prediction	7
1.2	Proposed System Objectives	9
3.1	End-to-End System Architecture of Hybrid Agricultural Price Forecasting System	23
3.2	Hybrid Forecasting Architecture	31
4.1	Data Sources Module	36
4.2	Data Ingestion and Storage Module	37
4.3	Processing and Analytics Module	37
4.4	Forecasting Module	38
4.5	Web Application Module and User Modules	39
4.6	Non-Functional Module	40
4.7	Landing Page Interface	40
4.8	Price Intelligence Overview	41
4.9	Price Forecasting Dashboard Interface	41
4.10	Crop, Market and Seasonal Selection Module	41
4.11	Forecast Horizon Selection Module	42
4.12	Predicted Average Price Summary (Long-Term Forecast)	42
4.13	Price Trend Analysis	42
4.14	Overall Price Trend Analysis	43
4.15	Listing of Crops for Price Trends	43
4.16	Wheat Price Overview	44
4.17	Gram Dal Price Overview	44
Figure No.	Title	Page No.

4.18	Bajra Price Overview	44
4.19	Soyabean Price Overview	44
4.20	Month-wise Forecast Trends	45
4.21	Year-wise Forecast Trend Graph	45
5.1	Performance Comparison of Existing and Proposed Models	48
5.2	Error Metric Comparison Graph	48
5.3	Model Stability Comparison Graph	49
5.4	Seasonality Pattern Comparison Graph	50
5.5	Market Trend Detection Accuracy Comparison Graph	51
5.6	Actual vs Forecasted Price Trend Using Hybrid RF-LSTM Model	51

LIST OF TABLES

Table No.	Title	Page No.
1.1	Descriptive Statistics of Collected Commodity Price Data	7
1.2	Comparative Performance of Forecasting Models on Price Prediction	8
1.3	Importance of Price Forecasting for Different Stakeholders	10
3.1	Dataset Description	28
3.2	Preprocessing Techniques	29
3.3	Feature Categories	30
3.4	Existing System vs Proposed Work	34
4.1	Implementation Challenges and Solutions	46
5.1	Error Metric Comparison Between Existing and Proposed Models	48
5.2	Comparison of Individual and Hybrid Models	49
5.3	Model Stability Comparison Between Existing and Proposed Models	49
5.4	Seasonality Pattern Comparison Between Existing and Proposed Models	50
5.5	Market Trend Detection Accuracy Comparison	50
5.6	Requirement-to-Outcome Mapping	52
5.7	Impact Analysis of the Proposed System	52

SYMBOLS & ABBREVIATIONS

Acronym	Expansion
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
ARIMA	Auto-Regressive Integrated Moving Average
LSTM	Long Short-Term Memory
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
TDNN	Time-Delay Neural Network
VAR	Vector Auto Regression
XGBoost	Extreme Gradient Boosting
CNN	Convolutional Neural Network
TCN	Temporal Convolutional Network
GRU	Gated Recurrent Unit
TFT	Temporal Fusion Transformer
ETL	Extract, Transform, Load
FCI	Food Corporation of India
GDP	Gross Domestic Product

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Agriculture has always been the backbone of the Indian economy, contributing significantly to national GDP and rural livelihoods. With nearly 60% of the population dependent on agriculture, the stability of food prices plays a critical role in ensuring food security and sustainable development. Among agricultural commodities, pulses and vegetables such as onion, potato, and tomato are considered essential due to their nutritional value and widespread consumption.

However, these commodities are highly prone to price volatility, driven by seasonal cycles, production variability, storage limitations, transportation inefficiencies, and unpredictable weather conditions. This volatility adversely affects farmers, traders, consumers, and policymakers alike.

The Department of Consumer Affairs currently monitors the daily prices of 22 essential food commodities through a network of 550 reporting centers across India. To stabilize market conditions, the government maintains buffer stocks of key commodities like pulses and onions. Despite these interventions, the unpredictability of prices remains a significant issue due to factors such as:

- Seasonal fluctuations in production
- Crop yield variability caused by weather conditions
- Market demand and supply imbalances
- Transportation and storage inefficiencies
- Policy interventions and global trade influences

With the rise of Artificial Intelligence and Machine Learning, there is an opportunity to develop advanced, data-driven models that can predict prices more accurately, helping farmers plan production, policymakers regulate supply, and consumers access affordable food. Agriculture is not only a source of livelihood for a majority of Indians but also a key sector that directly impacts food security, economic growth, and national stability. India, being one of the largest producers of pulses and vegetables globally, plays a significant role in the global food supply chain.

Despite this advantage, one of the critical challenges faced by the agricultural ecosystem is the instability of commodity prices. Vegetables like onion, potato, and tomato, along with pulses such as gram, tur, urad, moong, and masur, are consumed daily across all socioeconomic sections of the society. Even small fluctuations in their prices can lead to large-scale inflationary impacts, directly affecting household budgets and policymaking. Hence, predicting and stabilizing these prices is of national importance.

1.1.1 Global Context

Globally, agricultural markets are becoming increasingly interconnected and volatile due to climate change, globalization, and shifting demand patterns. Studies from the Food and Agriculture Organization (FAO) indicate that extreme weather events and supply chain disruptions have been major drivers of food price spikes in the last two decades. Price prediction, therefore, is not just a local concern but a global research priority.

1.1.2 Indian Context

In India, price rise of onions or pulses has often led to political and economic repercussions. For example, onion shortages have historically contributed to inflation spikes and have influenced government decisions on exports and imports. This demonstrates how essential commodities are directly linked to policy, economy, and public welfare.

1.2 PROBLEM STATEMENT

The agricultural sector in India forms the backbone of the nation's economy and food security, yet it remains highly vulnerable to the challenge of unpredictable price fluctuations in essential Agri-horticultural commodities such as pulses and vegetables (notably onion and potato).

Price instability in these commodities is not just an economic concern but a social one, as it directly impacts farmers' livelihoods, consumer affordability, and government policy decisions. Even minor variations in weather conditions, pest attacks, transportation bottlenecks, or shifts in market demand can lead to sudden spikes or crashes in commodity prices. This volatility has severe ripple effects: farmers often fail to secure fair returns for their produce, consumers struggle with inflated costs of daily essentials, and policymakers face mounting pressure to stabilize markets and ensure food availability.

Currently, the Department of Consumer Affairs attempts to mitigate these challenges by monitoring daily prices across multiple reporting centers, maintaining buffer stocks, and implementing interventions. However, these measures remain largely reactive in nature and often fall short of preventing crises. Traditional statistical models, such as ARIMA, though historically useful, are inadequate in today's context. These models struggle to capture the non-linear, multi-factor, and seasonal characteristics of agricultural price movements, especially in a country as diverse and complex as India. They also fail to integrate diverse data streams such as weather forecasts, soil health data, transportation costs, and global commodity trends, which play a significant role in determining prices.

The lack of robust and proactive forecasting mechanisms not only perpetuates uncertainty but also undermines long-term planning for all stakeholders. Farmers are unable to make informed sowing and selling decisions, leading to either surplus production and wastage or shortages that drive inflation. Consumers bear the brunt of rising costs of food items, which form a large proportion of household expenditure, especially in lower-income groups. For policymakers, inaccurate or delayed forecasts make it difficult to design effective procurement, subsidy, and buffer stock strategies, thereby affecting national food security and economic stability.

In this context, there is an urgent need to leverage advanced technologies, particularly Artificial Intelligence (AI) and Machine Learning (ML), to build predictive models capable of capturing the complex interplay of factors influencing agricultural prices. Unlike traditional methods, AI-ML based models can process vast amounts of heterogeneous data, identify hidden patterns, and continuously learn from new inputs. By integrating historical price trends, market intelligence, weather and climate variables, production levels, transportation and logistics costs, and even global market indicators, these models can generate more accurate, timely, and actionable forecasts. Such forecasts would empower farmers to plan cultivation cycles effectively, enable consumers to anticipate market conditions, and equip policymakers with data-driven insights to implement proactive interventions rather than reactive measures. Therefore, the core problem addressed in this study is the inadequacy of existing forecasting mechanisms in capturing the highly dynamic and multi-factorial nature of agricultural commodity prices in India.

Developing AI-ML based predictive systems that provide reliable price forecasts is essential for ensuring price stability, protecting farmer incomes, safeguarding consumer interests, and ultimately strengthening the nation's food security framework.

1.2.1 Our Understanding of the Problem

Agricultural commodity prices in India, especially for pulses and vegetables like onion and potato, are highly volatile and unpredictable. This volatility arises from a combination of natural, infrastructural, and policy-related factors. Farmers, consumers, and policymakers are all adversely impacted due to this instability, and existing forecasting methods fail to capture the real complexity of these markets.

Key factors contributing to the problem:

- **Weather Variability:** Unpredictable monsoons, floods, droughts, and heatwaves affect both yield and quality of crops.
- **Production Patterns:** Fluctuations in acreage and shifting cultivation preferences create unpredictable supply levels.
- **Storage Limitations:** Lack of adequate cold storage and warehousing infrastructure leads to wastage during harvest peaks and shortages later.
- **Transport Disruptions:** Poor rural connectivity, strikes, or rising fuel prices often block smooth supply chains.
- **Policy Interventions:** Export bans, MSP announcements, and sudden buffer stock releases cause abrupt market swings.
- **Traditional Forecasting Gaps:** Models like ARIMA can only capture simple trends and seasonality, but fail with non-linear, multi-factor, real-world data.

Impact on stakeholders:

- **Farmers:** Cannot make informed sowing/selling decisions, often forced into distress sales.
- **Consumers:** Suffer from sudden price hikes in essential daily food items.
- **Policymakers:** Struggle to stabilize prices without accurate forecasts, often reacting too late.

1.2.2 Planned Approach

1.2.2.1 Data Collection & Integration

- Gather historical price data of key commodities.
- Include exogenous factors: weather data, production stats, global trade inputs, and buffer stock information.
- Use multiple sources: government databases, open data APIs, and market intelligence reports.

1.2.2.2 Model Development (AI/ML based)

- Apply time-series models enhanced with external features.
- Use ML/DL methods (Random Forests, Gradient Boosting, LSTMs) to capture non-linear, seasonal, and volatile patterns.
- Continuously retrain models with new incoming data for adaptive forecasting.

1.2.2.3 Forecast Generation

- **Short-term forecasts (7–14 days):** For immediate planning in procurement and logistics.
- **Medium-term forecasts (1–3 months):** For policymaking, buffer stock planning, and trade strategies.
- **Long-term forecasts (6–12 months):** For strategic agricultural planning, infrastructure development, crop diversification, and long-term policy formulation.

1.2.2.4 Output Delivery & Visualization

- Interactive dashboards with price trend visualization.
- Real-time alerts (SMS, app notifications) for farmers and stakeholders.
- Scenario-based simulations to test “what-if” conditions for policymakers.

1.2.2.5 User-Centric Framework

- Multilingual, mobile-friendly design for farmer accessibility.
- Community-driven UX to ensure adoption in rural and urban settings alike.

1.3 AGRICULTURAL PRICE VOLATILITY

Price fluctuations in agricultural commodities are influenced by several interconnected factors:

- ❖ **Seasonality:** Many crops are harvested at specific times, creating periods of surplus and shortage.
- ❖ **Weather & Climate:** Unpredictable monsoons, floods, or droughts directly affect yield.
- ❖ **Storage & Supply Chain:** Lack of cold storage and inefficient logistics worsen the mismatch between supply and demand.
- ❖ **Government Interventions:** Imports, exports, subsidies, and buffer stock releases impact short-term price movements.

1.4 OBJECTIVES

The primary aim of this project is to develop an AI-ML based predictive model for forecasting the prices of essential agri-horticultural commodities such as pulses and vegetables (onion, potato, etc.). These objectives are designed to address the limitations of existing systems and provide accurate, reliable, and real-time insights for decision-makers.

1.4.1 General Objective

To design and implement a robust AI-ML framework capable of forecasting short-term and long-term prices of key agricultural commodities with high accuracy.

1.4.2 Specific Objectives

1.4.2.1 Data Collection and Preprocessing

- ❖ To gather historical price data from government sources, wholesale markets, and consumer affairs reports.
- ❖ To preprocess and clean the dataset for handling missing values and outliers.
- ❖ To integrate supplementary data such as weather conditions and production estimates to enrich the dataset.
- ❖ To perform data normalization and temporal alignment to ensure consistency across multiple data sources.

Table 1.1 Descriptive Statistics of Collected Commodity Price Data

Commodity	Data Range	Total Records	Missing Values (%)	Sources
Onion	2010 - 2023	4800	2.3	Govt, Mandis
Potato	2011 - 2023	4650	1.8	Govt, FCI
Gram Dal	2012 - 2023	4500	3.1	Govt Reports
Urad Dal	2011 - 2023	4700	2.7	Mandis, FCI

1.4.2.2 Feature Engineering

- To identify and incorporate key influencing factors such as seasonality, rainfall, buffer stock levels, imports/exports, and demand-supply gaps.
- To transform raw data into meaningful inputs for machine learning algorithms.

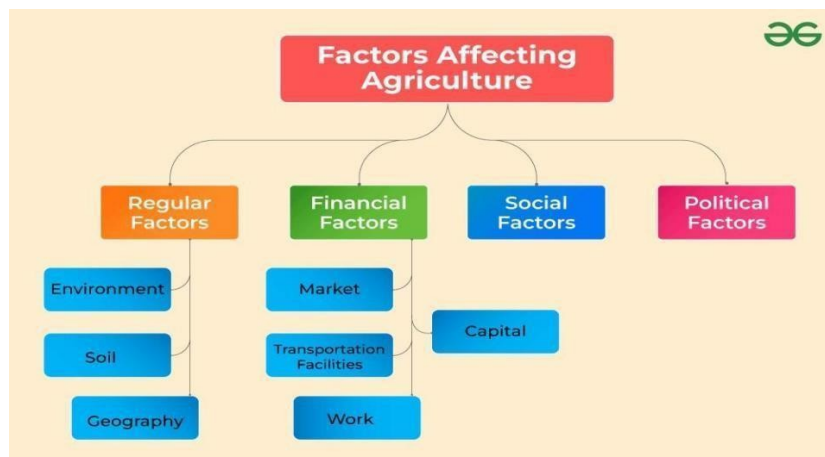


Figure 1.1 Influencing Factors for Price Prediction

1.4.2.3 Model Development and Training

- To experiment with multiple algorithms such as ARIMA, LSTM, Random Forest, and Prophet.
- To train and validate models on historical datasets, ensuring high predictive performance.
- To optimize hyperparameters for better generalization and reduce error rates.

1.4.2.4 Evaluation and Validation

- To measure the accuracy of predictions using evaluation metrics such as RMSE, MAE, and MAPE.

- To benchmark the developed AI-ML models against traditional forecasting techniques like ARIMA.

Table 1.2 Comparative Performance of Forecasting Models on Price Prediction

Model	RMSE	MAE	MAPE (%)	Remarks
ARIMA	15.6	12.1	18.5	Limited for seasonality
Random Forest	11.2	9.4	13.2	Good for nonlinear trends
LSTM	8.9	7.5	10.1	Best for sequential data
Prophet	10.3	8.9	12.4	Strong for trend capture

1.4.2.5 Web Application Overview

The final objective is to provide stakeholders with a user-friendly web application that translates complex forecasting outputs into simple, actionable insights. The application will have a clean and intuitive interface, structured around:

- ❖ **Dashboard Home:** A summary view displaying current prices, recent trends, and forecast highlights.
- ❖ **Interactive Charts & Graphs:** Visual representation of price forecasts (short-term and medium-term) with filters for commodity, region, and time horizon.
- ❖ **Scenario Analysis Module:** “What-if” exploration (e.g., effect of rainfall deficit or buffer stock release on prices).
- ❖ **Stakeholder Views:** Farmers receive a simplified price outlook with short-term (1–2 week) trend predictions to support planning decisions. Consumers are provided with retail price alerts and affordability indicators to assist in informed purchasing.
- ❖ **Notifications & Alerts:** Automated updates via dashboard widgets and optional SMS/Email integration.

This structure ensures that the system is not only analytically robust but also **accessible, scalable, and actionable**, serving the needs of diverse users from farmers to government decision-makers.

1.4.2.6 Deployment and Practical Use

- ❖ To build a user-friendly system/dashboard for real-time visualization of price forecasts.
- ❖ To assist policymakers in buffer stock management and farmers in crop planning.
- ❖ To make forecasting results accessible to stakeholders via web applications.

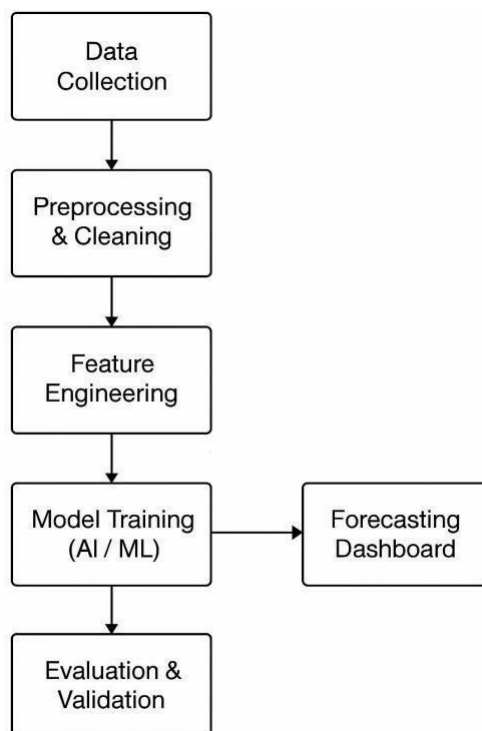


Figure 1.2 Proposed System Objectives

1.4.2.7 Expected Outcomes

- ❖ Reliable price predictions with improved accuracy compared to existing systems.
- ❖ Better decision-making support for government agencies, farmers, and consumers.
- ❖ Reduced price volatility through timely interventions and informed market strategies.

1.5 IMPORTANCE OF PRICE PREDICTION IN AGRICULTURE

Accurate price forecasting of agri-horticultural commodities such as pulses and vegetables is essential because it benefits multiple stakeholders:

Table 1.3 Importance of Price Forecasting for Different Stakeholders

Stakeholder	Importance
Farmers	Enables better crop planning, reduces losses, improves income predictability
Consumers	Ensures availability of affordable essential commodities
Policymakers	Helps design effective interventions like buffer stock release, import/export
Traders	Improves procurement, storage, and distribution strategies
Economy	Strengthens food security and stabilizes inflation

By enabling evidence-based decision-making, price forecasting supports national goals of food security and rural development.

1.6 ROLE OF AI AND ML IN PRICE PREDICTION

The complexity of agricultural price prediction demands methods that can handle large-scale, diverse, and often noisy datasets. Traditional models (e.g., ARIMA, exponential smoothing) rely heavily on historical prices and assume linearity. However, agricultural systems are non-linear, seasonal, and influenced by multi- dimensional factors. AI and ML techniques offer several advantages:

- ❖ **Data Integration:** AI models can incorporate diverse datasets, including weather, soil health, market demand, and policy changes.
- ❖ **Pattern Recognition:** Machine learning algorithms identify hidden patterns that humans or traditional models may overlook.
- ❖ **Adaptability:** Models can update themselves dynamically as new data is fed.
- ❖ **Predictive Accuracy:** Advanced methods such as LSTM neural networks are specifically designed for time-series data like commodity prices.

- ❖ **Decision Support:** AI-based tools can provide not only forecasts but also insights on why a price fluctuation is expected (feature importance).

1.7 SIGNIFICANCE

The importance of this project lies in its ability to provide real-world impact:

- ❖ Farmers gain decision support for crop selection and timing.
- ❖ Policymakers can avoid panic-driven interventions and design evidence-based strategies.
- ❖ Consumers are safeguarded against sudden price surges.
- ❖ The agricultural economy moves towards a data-driven ecosystem.

1.8 SCOPE

The primary goals of this project include:

- ❖ To analyze the historical price trends of pulses and vegetables across different markets.
- ❖ To identify major influencing factors such as seasonality, buffer stock release, crop production, and weather conditions.
- ❖ To develop and implement AI/ML models (Random Forest, LSTM, Ensemble Models) for price prediction.
- ❖ To compare the performance of AI/ML models with traditional approaches like ARIMA.
- ❖ To provide a decision-support system that aids farmers, policymakers, and traders.
- ❖ To forecast the price of pulses and vegetables (onion, potato, tomato, etc.) using AI/ML models with data collected from reporting centers.

1.9 LIMITATIONS OF THE EXISTING SYSTEM

Despite continuous monitoring and forecasting efforts, existing agricultural price prediction systems face several challenges that limit their effectiveness in real-world scenarios. These limitations highlight the need for advanced, data-driven approaches capable of handling complex market dynamics and supporting informed decision-making.

- Existing agricultural price forecasting systems primarily rely on traditional statistical models such as ARIMA, which are limited in capturing nonlinear price behavior and complex market dynamics.
- Most systems depend heavily on historical price data and fail to incorporate multidimensional factors such as weather conditions, production estimates, and market intelligence.
- Conventional approaches struggle to adapt to sudden market disruptions caused by supply-chain issues, policy interventions, or climatic variations.
- Limited scalability and lack of real-time decision support restrict their practical usability for farmers and policymakers.
- Existing systems often provide static forecasts without interactive visualization or user-specific insights, reducing their effectiveness in real-world applications.

1.10 OVERVIEW OF THE PROPOSED SYSTEM

The proposed system is designed to overcome the limitations of existing approaches by leveraging advanced AI and machine learning techniques. It provides a comprehensive, data-driven forecasting framework that integrates multiple influencing factors to deliver accurate and actionable price predictions.

- Utilizes AI and machine learning techniques to accurately forecast agricultural commodity prices.
- Integrates multi-source data including historical prices, weather conditions, production estimates, and market intelligence.
- Employs a hybrid forecasting approach combining LSTM and Random Forest models for improved accuracy and stability.
- Provides interactive visualizations and a web-based dashboard for easy interpretation of results.
- Supports timely decision-making for farmers, consumers, and policymakers through actionable insights.

In summary, the volatility of agricultural commodity prices continue to pose challenges for farmers, consumers, policymakers, and the overall economy. By leveraging historical data, seasonal patterns, market dynamics, and external factors, this project aims to design a robust forecasting framework that not only enhances prediction accuracy but also supports informed decision-making across the agricultural ecosystem.

CHAPTER 2

LITERATURE REVIEW

This chapter provides an overview of significant research carried out in the domain of agricultural commodity price forecasting. Various statistical, machine learning, and deep learning approaches have been explored by researchers worldwide to improve accuracy in predicting market prices. The following review highlights some of the most relevant and original works, detailing their methodology, findings, and limitations, while also positioning how our proposed system builds upon them.

2.1 EXISTING RESEARCH

2.1.1 Forecasting International Maize Prices Using Long Short-Term Memory Networks – Ronit Jaiswal, Girish K. Jha, Kapil Choudhary, Rajeev Ranjan Kumar (2023)

In 2023, Ronit Jaiswal, Girish K. Jha, Kapil Choudhary, and Rajeev Ranjan Kumar applied Long Short-Term Memory (LSTM) networks to forecast international monthly maize prices. Their study highlighted how LSTM models were able to capture both short-term fluctuations and long-term patterns far more effectively than conventional statistical approaches.

The results demonstrated superior accuracy and robustness, making LSTM a promising technique for commodity price forecasting. However, the work was largely restricted to a single commodity and lacked multivariate input factors, limiting its general applicability across diverse agricultural products.

2.1.2 Comparative Analysis of ARIMA, Time-Delay Neural Networks, and LSTM Models for Forecasting Indian Commodity Spot Prices – Manogna R. L., Aswini K. Mishra (2021)

Manogna R. L. and Aswini K. Mishra (2021) carried out a comparative study using ARIMA, Time-Delay Neural Networks (TDNN), and LSTM models for forecasting daily spot prices of Indian commodities such as cottonseed, castorseed, rapeseed, soybean, and guar seed. Their findings revealed that LSTM consistently achieved higher directional accuracy than ARIMA and TDNN, indicating its superiority in capturing price variations.

While this work emphasized the strength of deep learning approaches in Indian agricultural markets, the study was still limited in scope to a few commodities and primarily focused on short-term forecasting.

2.1.3 Machine Learning Approaches for Forecasting Wholesale and Retail Prices of Onion, Potato, and Tomato in India – H. P. Suresha, Mutturaj Uppaladinni, Krishna Kumar Tiwari (2021)

A notable study in 2021 by H. P. Suresha, Mutturaj Uppaladinni, and Krishna Kumar Tiwari focused on forecasting wholesale and retail prices of onion, potato, and tomato in India, while also incorporating variables such as diesel prices and arrival volumes. They compared multiple techniques including Auto ARIMA, RNN, LSTM, VAR, Random Forest Regression, and XGBoost.

The research found that machine learning approaches such as LSTM and XGBoost consistently outperformed traditional statistical models, demonstrating their ability to handle nonlinear patterns in agricultural data. Despite these promising results, the study was limited to selected commodities and did not extend its findings to pulses and other vegetables of national importance.

2.1.4 Deep Learning Models for Forecasting Weekly Onion Prices in Major Indian Markets – A. Kowsalya, G. Sridevi (2020)

In 2020, A. Kowsalya and G. Sridevi applied several deep learning models, including LSTM with attention, CNN, hybrid CNN-LSTM, and Temporal Convolutional Networks (TCN), to forecast weekly onion prices in major Indian markets such as Delhi, Mumbai, and Bangalore. Their study compared these approaches against conventional techniques such as GARCH, TDNN, and Support Vector Regression (SVR).

The results demonstrated that deep learning models, particularly hybrid ones, offered significantly higher accuracy and stability in volatile markets like onions. Nevertheless, the research was limited to a single commodity and did not incorporate broader market or environmental factors that may influence prices.

2.1.5 Application of LSTM Variants for Agricultural Commodity Price Forecasting: A Study on Rice, Wheat, Gram, Banana, and Groundnut – Murugesan, Mishra, Krishnan (2022)

Murugesan, Mishra, and Krishnan (2022) conducted an extensive study applying five different LSTM variants — vanilla LSTM, bi-LSTM, stacked LSTM, CNN- LSTM, and Conv-LSTM — to forecast price data for rice, wheat, gram, banana, and groundnut spanning 2000 to 2020.

The study revealed that all the LSTM-based models achieved accurate short-term forecasts with very low error metrics such as MAE, MAPE, and MSE. This highlighted the effectiveness of deep learning techniques in agricultural price prediction. However, the research primarily focused on univariate forecasting, without integrating additional features like weather conditions, trade policies, or demand-supply shocks.

2.1.6 A Hybrid Approach for Forecasting Agricultural Commodity Prices Using ARIMA, LSTM, and Random Forest – S. Ray, A. Lama, P. Mishra, T. Biswas, S. S. Das, B. Gurung (2023)

More recently, in 2023, S. Ray, A. Lama, P. Mishra, T. Biswas, S. S. Das, and B. Gurung proposed a hybrid forecasting approach combining ARIMA, LSTM, and Random Forest to model volatile agricultural commodity prices. In this framework, Random Forest was used for selecting the most relevant lag features, which were then utilized by ARIMA and LSTM to generate predictions. Their results showed that this hybrid architecture outperformed standalone models, particularly in handling high volatility and irregular price series. The limitation of this study was that it was tested on a narrow dataset, making it less generalizable for a wide variety of agricultural commodities.

From the reviewed works, it is evident that deep learning models, particularly LSTM and its variants, consistently outperform traditional statistical approaches in capturing the non-linear and volatile nature of agricultural commodity prices. However, most existing studies are constrained either by focusing on a limited set of commodities or by excluding important external factors such as weather, policy, or transportation costs. Our project addresses these gaps by proposing a multivariate, hybrid AI-ML framework tailored specifically to pulses and vegetables, aiming to enhance forecasting accuracy and reliability.

2.2 KEY OBSERVATIONS FROM EXISTING RESEARCH

2.2.1 Traditional Statistical Models

- ❖ Early research relied heavily on linear and time-series models like ARIMA, Exponential Smoothing, and regression-based approaches.
- ❖ These models were effective for short-term predictions under stable conditions but struggled with high volatility, seasonality, and multi-factor influences.

2.2.2 Machine Learning Approaches

- ❖ Recent studies increasingly employ supervised ML algorithms such as Random Forests, Gradient Boosting, and Support Vector Machines to predict commodity prices.
- ❖ These approaches can handle non-linear relationships, incorporate multiple features like weather, production data, and trade information, and adapt to changing market conditions.

2.2.3 Deep Learning & Hybrid Models

- ❖ Deep learning methods, including LSTMs and GRUs, have shown strong performance in capturing sequential dependencies and complex seasonal patterns.
- ❖ Hybrid models combining time-series analysis with ML or deep learning are emerging as effective solutions for multi- horizon forecasting.

2.2.4 Incorporation of External Factors

- ❖ Studies highlight the importance of including exogenous variables: rainfall, temperature, soil health, transportation/storage constraints, and government policies.
- ❖ Models incorporating these factors generally outperform those using historical prices alone, particularly for long-term and medium-term forecasts.

2.2.5 Decision Support & User-Centric Applications

- ❖ Some research has focused on delivering predictions via dashboards, mobile apps, and alert systems for farmers and policymakers.
- ❖ Explainable AI models are gaining attention to ensure trust and adoption among

non-technical users.

2.3 GAPS IN EXISTING KNOWLEDGE

Although significant progress has been made in agricultural commodity price forecasting using statistical, machine learning, and deep learning approaches, several critical gaps remain in existing research. Most studies focus on a **limited set of commodities**, often analyzing a single crop or a small group of products, which restricts the generalizability of the proposed models across diverse agricultural markets.

A major limitation observed is the **predominant reliance on univariate or partially multivariate models**, where forecasting is primarily based on historical price data. While some studies incorporate external variables, comprehensive integration of **multi-source factors** such as weather conditions, production estimates, market arrivals, buffer stock interventions, and policy-related variables is still limited. This reduces the ability of existing models to fully capture real-world market dynamics.

Furthermore, many deep learning-based approaches, particularly LSTM and its variants, emphasize prediction accuracy but offer **limited interpretability**, making them less suitable for policy-driven decision support systems. The lack of explainable insights hinders adoption by policymakers and other non-technical stakeholders who require transparency in forecasting outputs.

Another notable gap is the **insufficient focus on hybrid modeling frameworks** that effectively combine temporal learning capabilities of deep learning models with the feature interpretability and robustness of machine learning techniques. While some hybrid models have shown promise, their application is often constrained by narrow datasets and limited validation across varying market conditions.

Finally, relatively few studies address the **deployment and usability aspects** of forecasting systems. Many works stop at model development without providing scalable, user-centric platforms such as dashboards or alert systems that translate predictions into actionable insights. This gap limits the practical applicability of research outcomes in real-world agricultural market management. The proposed work aims to bridge these gaps by developing a **multivariate, hybrid AI–ML forecasting framework** for essential pulses and vegetables, integrating diverse external factors and delivering predictions through an interactive, decision-support-oriented web platform.

2.4 VALUE ADDITION OF THE PROPOSED SYSTEM

The proposed project adds significant value to existing agricultural commodity price forecasting systems by addressing critical limitations identified in prior research and practical implementations. Unlike traditional approaches that rely primarily on historical price trends, this system integrates multi-source data including weather conditions, production estimates, market intelligence, and buffer stock information, enabling a more comprehensive understanding of price dynamics.

The use of a hybrid AI–ML framework combining Long Short-Term Memory (LSTM) networks and Random Forest models enhances both prediction accuracy and robustness. While LSTM effectively captures temporal dependencies and seasonal patterns, Random Forest improves model interpretability and handles nonlinear relationships, resulting in more reliable forecasts under volatile market conditions.

Another key value addition lies in the system’s practical orientation. By deploying the forecasting models through a web-based dashboard with interactive visualizations, alerts, and role-based views, the project translates complex analytical results into actionable insights. This improves accessibility for farmers, consumers, and policymakers, bridging the gap between research and real-world decision-making.

Furthermore, the proposed system supports timely policy interventions such as buffer stock management and price stabilization measures. By enabling short-term and long-term price predictions, the framework assists stakeholders in proactive planning rather than reactive responses. In addition to improved forecasting accuracy and practical deployment, the proposed system is designed with scalability and adaptability in mind. The modular architecture allows seamless integration of additional commodities, regional markets, and real-time data streams as they become available. This flexibility ensures that the framework can evolve with changing agricultural and economic conditions.

Moreover, the inclusion of feature importance analysis and performance monitoring enhances transparency and trust in model predictions, making the system not only technically robust but also suitable for long-term implementation in agricultural market intelligence and policy planning. Overall, the project contributes a scalable, data-driven, and user-centric forecasting solution that enhances market transparency, supports food security, and promotes a sustainable agricultural ecosystem in India.

CHAPTER 3

METHODOLOGY

The present chapter provides a description of the structure, sources of data, steps of data preparation, ways in which to create models (features), and methods of forecasting used to create forecasts for agricultural commodity prices. The proposed solution can be described as a modular, scalable approach to forecasting, that encompasses all the steps required to develop and evaluate forecasting products.

3.1 SYSTEM OVERVIEW

This research focuses on creating accurate price forecasts for the primary agricultural commodities of onion, potato, and tomato using a new predictive methodology that incorporates the following:

- ❖ Historical price data
- ❖ Additional time series price data
- ❖ Real-time data (e.g., weather and crop status)
- ❖ Machine learning and deep learning models
- ❖ Visualization interface

The proposed methodology has the following five high-level components:

- ❖ Data collection
- ❖ Data preparation
- ❖ Feature generation
- ❖ Prediction model development
- ❖ Evaluation and presentation of predictions

The overall system is designed as a modular, end-to-end forecasting framework in which each component operates independently while remaining seamlessly integrated with the others. Data flows sequentially from collection and preparation stages to feature generation and predictive modeling, and finally to evaluation and visualization. This structured architecture ensures scalability, robustness, and ease of interpretation, enabling the system to deliver reliable price forecasts and actionable insights for various stakeholders. The modular design also allows future extensions such as the inclusion of additional commodities and real-time data sources.

3.2 PROBLEM-SOLVING APPROACH

The proposed solution follows a data-driven forecasting approach using artificial intelligence and machine learning techniques. The methodology begins with collecting multi-source agricultural data, followed by data preprocessing and feature engineering to enhance data quality. Machine learning and deep learning models are then trained to forecast commodity prices. A hybrid modeling strategy combining Random Forest and Long Short-Term Memory (LSTM) networks is adopted to improve prediction accuracy and robustness. Finally, the results are evaluated using standard performance metrics and visualized through a web-based dashboard to support decision-making.

3.3 TECHNOLOGIES USED

This project employs a combination of modern data science, machine learning, deep learning, and web development technologies to design and implement an AI-ML based agricultural commodity price forecasting system. The selected technologies support efficient data processing, accurate prediction, and user-friendly visualization of results.

3.3.1 Programming Language

3.3.1.1 Python

- Python is used as the primary programming language due to its simplicity, extensive library support, and strong ecosystem for data science and machine learning.
- It enables rapid development of data preprocessing pipelines, model implementation, and result analysis.
- Python's compatibility with web frameworks allows seamless integration between backend logic and the frontend interface.

3.3.2 Data Processing and Analytics Libraries

3.3.2.1 Numpy and Pandas

- NumPy is used for numerical computations and efficient array operations.
- Pandas is employed for handling structured datasets, time-series manipulation, missing value treatment, and exploratory data analysis.

3.3.2.2 Matplotlib and Seaborn

- These libraries are used to visualize historical trends and forecast results.
- Graphical representations help in validating prediction accuracy.

3.3.3 Machine Learning Frameworks

3.3.3.1 Scikit-learn

- Scikit-learn is used to implement machine learning models such as Random Forest.
- It provides tools for data splitting, feature scaling, model evaluation, and performance metrics.
- Feature importance analysis using Random Forest helps interpret the contribution of different input variables.

3.3.4 Deep Learning Frameworks

3.3.4.1 TensorFlow and Keras

- TensorFlow, along with its high-level API Keras, is used to develop and train Long Short-Term Memory (LSTM) networks.
- LSTM models are effective for time-series forecasting as they capture long-term temporal dependencies and seasonality.
- Keras simplifies model construction, hyperparameter tuning, and training visualization.

3.3.5 Forecasting Models

3.3.5.1 Random Forest Model

- Used to model nonlinear relationships between features and commodity prices.
- Provides robustness against overfitting and allows interpretation of feature importance.

3.3.5.2 Long Short-Term Memory (LSTM) Model

- Applied for sequential price forecasting tasks.
- Capable of learning long-term dependencies and seasonal trends in agricultural price data.

3.3.5.3 Hybrid LSTM–Random Forest Model

- Combines outputs from both LSTM and Random Forest models.
- Improves prediction accuracy and stability under volatile market conditions.

3.3.6 Web Application and Visualization Technologies

3.3.6.1 Streamlit / Flask (Frontend & Backend Integration)

- Used to develop a web-based dashboard for presenting forecasting results.
- Enables interactive selection of commodities and visualization of predicted price trends.

3.3.6.2 HTML, CSS, and JavaScript

- Used to design and enhance the user interface for better usability and responsiveness.
- Supports role-based access and dynamic visualization components.

3.3.7 Data Storage and Management

3.3.7.1 CSV / Local Database

- Used for storing historical datasets and intermediate processed data.
- Enables easy data retrieval and experimentation during model development.

3.3.7.2 Cloud Storage

- Used for scalable storage and deployment.

3.3.8 Development Environment and Tools

3.3.8.1 Jupyter Notebook

- Used for data exploration, preprocessing, model training, and experimentation.
- Provides an interactive environment for iterative development and analysis.

3.3.8.2 Visual Studio Code / PyCharm

- Used for code development, debugging, and project management.

3.3.9 Version Control

3.3.9.1 Git and GitHub

- Used for version control and collaboration.
- Helps track code changes, manage different versions, and maintain project documentation.

3.3.10 Evaluation Metrics and Tools

3.3.10.1 Performance Metrics (MAE, RMSE, MAPE)

- Used to evaluate forecasting accuracy and model reliability.
- Helps compare different models and validate improvements achieved by the hybrid approach.

The integration of Python-based data processing, advanced machine learning frameworks, and web visualization technologies enables the development of a robust and scalable agricultural price forecasting system. These technologies ensure high accuracy, interpretability, and practical usability for real-world decision support.

3.4 SYSTEM ARCHITECTURE

The proposed system architecture is designed as a modular, scalable, and robust framework that seamlessly integrates multiple components, including data acquisition, preprocessing, modeling, and visualization modules, ensuring efficient data flow, high performance, and adaptability to evolving requirements. Furthermore, the architecture supports real-time data processing and continuous model updates, enhancing prediction accuracy and system responsiveness.

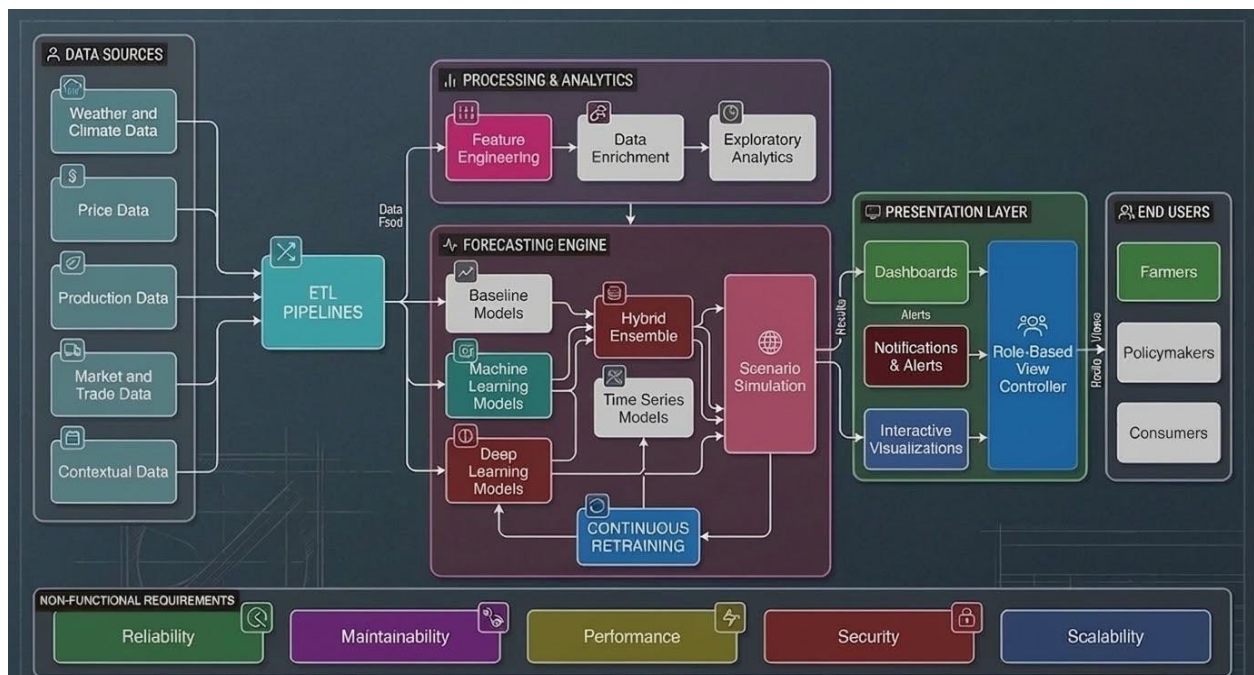


Figure 3.1 End-to-End System Architecture of Hybrid Agricultural Price Forecasting System

The above figure illustrates the end-to-end system architecture of the agricultural commodity price forecasting solution, depicting the systematic flow of data from multiple sources through various analytical and modeling stages to deliver meaningful insights to end users. The architecture integrates diverse data sources, efficient data ingestion mechanisms, advanced preprocessing and analytics modules, and hybrid AI–ML forecasting models to generate accurate price predictions. The processed outputs are delivered through a user-friendly web-based application, enabling interaction with different stakeholders such as farmers, policymakers, and analysts. The overall framework is organized into six major layers, namely **Data Sources, Data Ingestion and Storage, Processing and Analytics, Forecasting Engine, Presentation Layer, and User Layer**, ensuring modularity, scalability, and seamless interaction among system components.

3.4.1 Data Sources Layer

The accuracy and robustness of forecasts depend on the richness and diversity of input data. Our system integrates multiple structured and unstructured sources, ensuring both historical and real-time signals are captured.

- ❖ **Price Data:** Daily/weekly price records from Department of Consumer Affairs monitoring centers across regions.
- ❖ **Agricultural Production Data:** Crop sowing area, yield statistics, mandi arrivals, and procurement records.
- ❖ **Weather & Climate Data:** Rainfall, temperature, drought/flood indices, soil moisture, and long-range IMD forecasts.
- ❖ **Market & Trade Data:** Import/export volumes, buffer stock levels, logistics disruptions, and global commodity indices.
- ❖ **Contextual Data:** Festival periods, consumer demand spikes, news sentiment around crop shortages or bumper harvests.

By combining these inputs, the system captures both **direct and indirect factors** influencing commodity prices.

3.4.2 Data Ingestion and Storage Layer

A **data pipeline** ensures seamless ingestion, cleaning, and organization of raw data into a usable format.

3.4.2.1 ETL Pipeline (Extract, Transform, Load)

- ❖ Automated scripts fetch data from APIs, government reports, and other repositories.

- ❖ Raw data is cleaned (handling missing values, unit conversions, duplicate removal).
- ❖ Transformed into standardized formats for downstream processing.

3.4.2.2 Storage

- ❖ **Data Lake** for storing raw, semi-structured, and unstructured data.
- ❖ **Relational Database (SQL/NoSQL)** for processed, structured datasets ready for model training.
- ❖ **Scalability:** Cloud-based storage ensures data growth is easily managed, and real-time updates are supported.

3.4.3 Processing and Analytics Layer

This layer prepares data for modeling and generates intermediate analytics to support interpretation.

3.4.3.1 Feature Engineering

- ❖ Lag variables (7-day, 14-day, 30-day price trends).
- ❖ Seasonal indicators (harvest months, festivals).
- ❖ Weather-yield lags (rainfall impacting yield with 1–2 month delay).

3.4.3.2 Data Enrichment

Correlating price data with production, trade, and weather signals to enrich the dataset and capture broader market influences.

3.4.3.3 Exploratory Analytics

Visual dashboards to identify correlations, anomalies, and historical price cycles. This ensures the forecasting models are trained on **high-quality, enriched data** that captures both short-term signals and long-term patterns.

3.4.4 Forecasting Engine (AI/ML Layer)

At the heart of the architecture is the forecasting engine, which blends multiple models to provide robust predictions.

- ❖ **Baseline Models:** ARIMA, Prophet — used for benchmarking performance.
- ❖ **Machine Learning Models:** Gradient Boosting (XGBoost, LightGBM) for handling tabular + categorical data.

❖ **Deep Learning Models**

- **LSTM/GRU Networks:** For capturing long-term dependencies in time-series.
- **Temporal Fusion Transformers (TFT):** For multi- horizon forecasting using exogenous variables.

❖ **Hybrid Ensemble:** Combining outputs from multiple models to improve accuracy and reduce bias.

❖ **Scenario Simulation:** Ability to project outcomes under different assumptions (e.g., rainfall deficit, export ban).

The forecasting engine continuously **re-trains** on new data to adapt to changing market dynamics.

3.4.5 Presentation Layer

The forecasting outputs are delivered to end-users via a modern, intuitive web application.

- ❖ **Dashboard:** High-level overview of current prices, historical trends, and forecast ranges.
- ❖ **Interactive Visualizations:** Line charts, bar charts, and heatmaps for commodity and region comparisons.
- ❖ **Scenario Analysis Module:** Allows policymakers to simulate “what- if” scenarios (e.g., effect of releasing X buffer stock).
- ❖ **Role-based Views:** Farmers get a 7–14 day price outlook and consumers receive retail price alerts with budget indicators. Policymakers obtain medium-term insights for buffer stock and trade decisions.
- ❖ **Notifications & Alerts:** Email, SMS, or WhatsApp integration for real- time updates.

The frontend emphasizes **clarity, accessibility, and decision-support**, ensuring insights are both usable and actionable.

3.4.6 System Integration and Workflow

The proposed system architecture follows a structured, layered workflow that ensures efficient data acquisition, processing, analysis, and dissemination of forecasting results. Each layer is designed to perform specific functions while seamlessly interacting with other components, enabling smooth data flow and reliable system performance. This modular design improves scalability, maintainability, and adaptability to changing data sources and analytical requirements. The complete flow of the architecture can be summarized as follows:

3.4.6.1 Data Sources (Price, Weather, Production, Trade, Contextual)

This layer collects data from multiple government portals, market databases, meteorological agencies, and external information systems to ensure comprehensive coverage of factors influencing commodity prices.

3.4.6.2 Data Ingestion (ETL Pipelines → Data Lake + Database)

Raw data is extracted, transformed, and loaded through automated pipelines into centralized storage systems, ensuring consistency, integrity, and availability for downstream processing.

3.4.6.3 Processing & Feature Engineering (Lag Variables, Seasonality, Weather Correlations)

The collected data is cleaned, normalized, and transformed into meaningful features such as lag indicators, rolling statistics, seasonal indices, and weather correlation metrics to enhance model learning capability.

3.4.6.4 Forecasting Engine (ML/DL Ensemble + Scenario Modeling)

This layer applies hybrid machine learning and deep learning models, including Random Forest and LSTM networks, to generate accurate short-term and long-term price forecasts and evaluate alternative market scenarios.

3.4.6.5 Presentation Layer (Web Application + Alerts)

Forecast outputs are visualized through an interactive web-based dashboard and alert mechanisms, enabling users to explore trends, compare scenarios, and receive timely notifications.

3.4.6.6 End Users (Farmers, Policymakers, Consumers)

The final layer delivers customized insights to different stakeholders, supporting informed decision-making, market interventions, and strategic planning.

Overall, this layered architectural design ensures efficient integration of heterogeneous data sources, robust analytical processing, reliable forecasting, and effective communication of results. By combining automation, advanced analytics, and user-centric visualization, the proposed framework provides a comprehensive decision-support platform for agricultural commodity price management.

3.4.7 Non-Functional Considerations

- ❖ **Scalability:** Cloud-native design allows for expansion across new commodities and geographies.
- ❖ **Performance:** Real-time or near-real-time forecasting updates with low latency.
- ❖ **Maintainability:** Modular architecture supports easy updates and integration of new data sources/models.
- ❖ **Reliability:** Automated monitoring of model drift and fallback mechanisms to simpler models in case of anomalies.

In essence, the system architecture ensures that the forecasting framework is **comprehensive, resilient, and impactful**, laying the foundation for long-term food security and market stability.

3.5 DATA COLLECTION

Data is collected from reliable sources such as government agricultural portals, market information systems, and weather datasets. The collected data includes historical commodity prices, production estimates, and weather-related attributes. Multi-source agricultural data is used in the forecasting system to create a more reliable forecasting and more accurate forecast prediction. The datasets that are collected include the following:

- Commodity price historical records
- Production and yield type statistics
- Weather and climate data
- Market intelligence metrics
- Buffer stock and intervention information

Table 3.1 Dataset Description

Datatype	Description	Frequency	Purpose
Price Data	Daily commodity prices	Time-series	Target variable
Production Data	Crop yield & output estimates	Seasonal	Supply estimation
Weather Data	Rainfall, temperature	Daily	Climate impact
Market Data	Demand, arrivals, trends	Periodic	Market behavior
Buffer Stock	Govt. intervention data	Event-based	Policy influence

3.6 DATA PREPROCESSING

Data preprocessing involves handling missing values, removing inconsistencies, and normalizing values to ensure uniformity. Feature engineering techniques such as lag features, rolling averages, and seasonal indicators are applied to capture temporal patterns. Exploratory data analysis is performed to identify trends, seasonality, and price volatility. To ensure model reliability, preprocessing is performed through the following steps:

- ❖ Removal of duplicate records
- ❖ Handling missing values using interpolation or statistical imputation
- ❖ Outlier detection and smoothing
- ❖ Data normalization and scaling
- ❖ Temporal alignment of multi-source datasets

This step ensures consistency and improves convergence during training.

Table 3.2 Preprocessing Techniques

Tasks	Techniques Used	Purpose
Missing values	Mean / interpolation	Completeness
Noise handling	Smoothing	Stability
Normalization	Min–Max scaling	Faster convergence
Date alignment	Time indexing	Temporal consistency
Outlier handling	IQR / thresholding	Robust learning

3.7 FEATURE ENGINEERING

Feature engineering plays a critical role in capturing patterns hidden in agricultural time-series data. The features extracted are:

- ❖ Lagged price features ($t-1$, $t-7$, $t-30$)
- ❖ Moving averages and rolling statistics
- ❖ Seasonal indicators (month, week)

- ❖ Trend components
- ❖ Weather-based attributes
- ❖ Production and buffer-stock indicators

These features help models learn temporal dependencies and external influences.

Table 3.3 Feature Categories

Feature Category	Examples
Lag Features	Price($t-1$), Price($t-7$), Price($t-30$)
Seasonal Features	Month, Week, Seasonal index
Statistical Features	Moving average, variance
Weather Features	Rainfall, temperature
Market Indicators	Arrivals, buffer stock levels

3.8 FORECASTING MODELS AND EXPERIMENTAL SETUP

A combination of Machine Learning & Deep Learning techniques is used to develop a reliable forecasting tool. The experimental setup involves training and testing the forecasting models using historical agricultural data. The dataset is divided into training and testing subsets to validate model performance. For that reason, we have identified 3 Models that will be used in the development of this tool as follows:

3.8.1 Random Forest (RF) Model

The Random Forest Model is the baseline model for Machine Learning because of its robust capabilities to handle non-linear data relationships and to produce feature importance scores for interpretability.

3.8.2 Long Short-Term Memory (LSTM) Model

Long Short-Term Memory Networks (LSTM) are designed to capture long term temporal features and dependencies of prices. LSTM will help model the seasonality and sequential dependencies found in most agricultural related data.

3.8.3 Hybrid Models (LSTM+RF)

The predictions from Random Forest and LSTM will be combined into Hybrid Ensemble Forecasting Models to capitalize on the following:

- ❖ Temporal Learning from LSTM
- ❖ Feature based Learning from Random Forest

The final output of Hybrid Models will produce a more stable forecast accuracy. Model performance is assessed using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE). The experiments are conducted in an offline environment using Python-based frameworks.

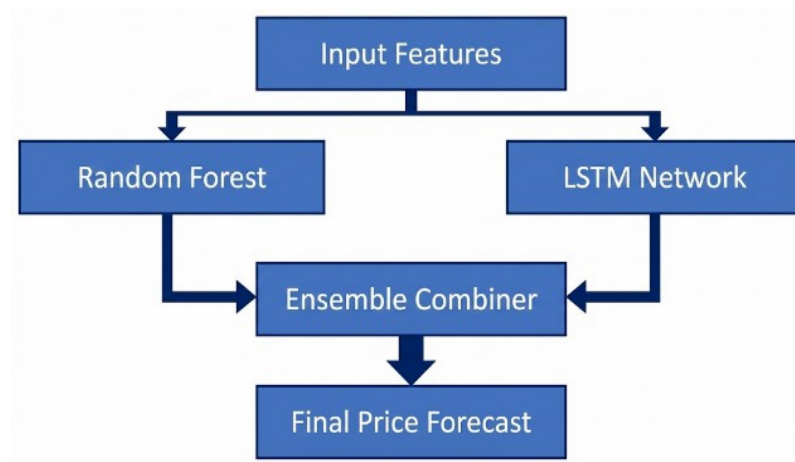


Figure 3.2 Hybrid Forecasting Architecture

3.9 MODEL TRAINING AND EVALUATION

The developed forecasting models were trained and evaluated using historical agricultural price data under a structured experimental setup. The training and evaluation process involves a series of steps.

3.9.1 Data Splitting

The collected dataset was divided into training and testing subsets to ensure an unbiased evaluation of the forecasting models. The training dataset was used to learn underlying patterns in agricultural price data, while the testing dataset was reserved for validating predictive performance on unseen data. This approach helps assess the generalization capability of the models under varying market conditions.

3.9.2 Model Training

Machine learning and deep learning models were trained using historical price data and engineered features. Random Forest models were trained to capture nonlinear relationships among input variables, while LSTM networks were trained to learn temporal dependencies and seasonal patterns. A hybrid model was developed by combining the outputs of both models to improve robustness and accuracy and thus contributing to effective planning and informed decision-making.

3.9.3 Model Evaluation

The trained models were evaluated using standard regression metrics commonly applied in time-series forecasting. **Mean Absolute Error (MAE)** was used to measure average prediction error, **Root Mean Square Error (RMSE)** was employed to penalize larger deviations and assess model stability, and **Mean Absolute Percentage Error (MAPE)** provided a percentage-based measure of prediction accuracy. The use of multiple metrics enabled a comprehensive comparison of model performance.

3.10 OUTPUT GENERATION AND VISUALIZATION

The forecasting outputs were converted into meaningful insights through an interactive and user-friendly visualization layer. The output generation process includes the following components:

3.10.1 Web-Based Dashboard

- ❖ Displays predicted prices along with historical trends.
- ❖ Allows users to select commodities, time horizons, and scenarios.

3.10.2 Interactive Visualizations

- ❖ Time-series plots for actual vs. predicted prices.
- ❖ Comparative views across different forecasting models.
- ❖ Trend and volatility indicators for better interpretation.

3.10.3 Decision-Support Features

- ❖ Scenario-based insights to evaluate potential market outcomes.
- ❖ Role-based access ensures relevant information for different stakeholders.

3.10.4 Usability and Interpretability

- ❖ Simplifies complex model outputs into visual formats.
- ❖ Enhances usability for policymakers, analysts, and decision-makers.

By translating complex analytical results into intuitive visual formats, the system supports informed decision-making and effective policy intervention.

3.11 EXISTING SYSTEM AND PROPOSED WORK

3.11.1 Existing System

Existing agricultural commodity price forecasting systems primarily rely on traditional statistical models such as ARIMA and related time-series techniques. These models are mainly dependent on historical price data and assume linear relationships within the dataset. While such approaches are effective under stable market conditions, they struggle to handle complex, nonlinear interactions and sudden market shocks caused by weather variability, supply-chain disruptions, or policy changes.

Moreover, traditional systems are limited in scope, often providing only short-term forecasts based on past trends and basic seasonality. The static nature of these models requires frequent manual recalibration, reducing their adaptability to rapidly changing market dynamics. As a result, existing systems offer moderate forecasting accuracy and limited decision-support capability for stakeholders.

3.11.2 Proposed Work

The proposed work introduces an AI–ML based forecasting framework designed to overcome the limitations of traditional approaches. By integrating diverse data sources such as historical prices, weather conditions, production estimates, logistics factors, and policy-related information, the system captures a comprehensive view of agricultural market dynamics. Machine learning and deep learning models, including Random Forest and LSTM, are employed to model nonlinear, multi-factor relationships and temporal dependencies.

Unlike static traditional models, the proposed framework supports multi-level forecasting, enabling short-term and medium-term price predictions to meet the needs of different stakeholders. The adaptive nature of the AI–ML models allows continuous learning and improved accuracy under volatile conditions.

Table 3.4 Existing System vs Proposed Work

Factor	Existing System (Traditional Models like ARIMA)	Proposed Work (AI-ML Based Models)
Data Sources	Limited to historical price data, mainly time-series trends	Integrates diverse datasets: historical prices, weather, production, logistics, global trade, and policy interventions
Modelling Capability	Captures only linear patterns, simple trends, and basic seasonality	Handles non-linear, multi-factor, and dynamic relationships using ML/DL models (RF, GBM, LSTM)
Forecast Accuracy	Moderate; fails under sudden shocks, high volatility, or multi-factor influences	Higher accuracy through feature-rich models, continuous learning, and adaptive retraining
Scope of Forecasting	Mostly short-term, limited to basic seasonal patterns	Multi-level: short-term (7–14 days), medium-term (1–3 months), adaptable to diverse needs
Adaptability	Static models; need manual updates and recalibration	Dynamic, self-learning models that update automatically with new data

In summary, the comparative analysis highlights the clear advantages of the proposed AI–ML based forecasting framework over traditional statistical models. By integrating multi-source data and advanced learning techniques, the system overcomes the limitations of linear and static forecasting approaches, resulting in more accurate, adaptive, and scalable price predictions. The hybrid RF–LSTM model effectively captures nonlinear patterns, seasonal trends, and market volatility.

Furthermore, the enhanced predictive capability supports proactive planning and informed decision-making for farmers, consumers, and policymakers. By enabling better anticipation of market fluctuations and timely interventions, the proposed framework contributes to improved market stability, greater transparency, and strengthened food security. It also reduces uncertainty in agricultural markets and helps stakeholders optimize resource allocation. Overall, the system fosters a more resilient, transparent, and data-driven agricultural ecosystem that supports sustainable market growth and informed decision-making.

CHAPTER 4

CONCLUSION

This project successfully addressed the persistent challenge of agricultural commodity price volatility in India by designing and implementing an AI-ML based forecasting framework for essential agri-horticultural commodities such as onion, potato, and tomato. The key findings of the study demonstrate that integrating multi-source data—including historical price trends, production estimates, weather indicators, and market-related variables—with advanced machine learning and deep learning techniques significantly enhances forecasting accuracy compared to traditional statistical models. The proposed hybrid Random Forest-LSTM approach consistently achieved lower error metrics such as MAE, RMSE, and MAPE, while also exhibiting strong capability in capturing nonlinear relationships, seasonal patterns, and short-term as well as long-term price fluctuations. Quantitative performance evaluation and visual trend analysis further validated the robustness, stability, and predictive effectiveness of the hybrid model under varying market conditions. The impact and importance of the project are evident in both its technical contribution and real-world applicability. By transforming complex forecasting outputs into intuitive, role-based insights through a web-based dashboard, the system bridges the gap between advanced analytics and practical decision-making. Farmers benefit from simplified short-term and medium-term price outlooks that support better production and sales planning, while policymakers gain access to reliable forecasts that assist in timely market interventions, buffer stock management, and price stabilization measures. The system also enhances market transparency, supports proactive policy formulation, and contributes to broader goals of food security and economic stability. This project highlights the growing potential of AI-driven decision support systems in strengthening agricultural market intelligence and enabling data-driven governance. Looking ahead, the proposed framework offers substantial scope for further enhancement and expansion. Future work may involve extending the system to cover additional agricultural commodities and wider geographical regions, including state-level and national-level forecasting. Integrating real-time data streams such as live APMC market feeds, satellite-based weather data, and logistics indicators could further improve prediction accuracy and responsiveness. Improving model interpretability through explainable AI techniques would help increase stakeholder trust and adoption, particularly among non-technical users. With continued refinement, the proposed approach has the potential to play a significant role in building a resilient, data-driven, and future-ready agricultural ecosystem in India.

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